Canvas Quiz 4, Modules 6–7a (graded)

Module 6 - Multiple Dosings · Canvas quiz · 5 Oct 2026

Name _____ Date _____

Question 1 Why are multiple dosage regimens are used?

A. to maintain plasma drug levels within the therapeutic window
B. to decrease the MEC
C. to increase the clearance of drugs
D. to decrease the rate of elimination of drug from the body

Question 2 What is the expected maximum steady-state concentration following IV bolus injections of 17 mg/kg to a 75 kg male every 8 hours? The drug has an elimination half-life of 4 hours, and the apparent volume of distribution is 15% of body weight. Give your answer in (mg/L), rounded to the nearest tenth (one decimal place) with no units.

Question 3 Which is the best option to increase the steady-state concentration in a dosing regimen?

A. decrease the dosing interval
B. decrease the elimination half-life
C. increase the clearance
D. increase the dosing interval

Question 4 Four hundred milligrams of an antibiotic was infused intravenously over a period of 2 hours. Eight hours after the start of the first infusion, a second 400-mg dose was infused again over a period of 2 hours. The half-life of the antibiotic is approximately 4 hours and the apparent volume of distribution is approximately 20 L. What is the plasma drug concentration 4 hours after the cessation of the second infusion? Give your answer in mg/L rounded to the nearest tenth (one decimal place) with no units.

Question 5 Which of the following results if the dose is increased on a medication that is to be administered orally at regularly repeating intervals?

A. Decreased clearance
B. Increased steady-state t_{max}
C. Decreased rate of absorption
D. Increased steady-state C_{max}

Question 6 A 74-kg patient was administered a 750 mg oral dose of a new drug every 8 hours. The drug has an oral bioavailability of 92%, apparent volume of distribution of 0.29 L/kg, elimination half-life of 5 hours, and the absorption half-life of 1.5 hour. When does the expected maximum concentration of drug in the plasma at steady-state occur? Give your answer in hours rounded to the nearest tenth (one decimal place) with no units.

Question 7 Which of the following best describes the assumptions associated with intermittent IV infusions?

A. intermittent IV infusions decrease elimination half-life which leads to increased efficacy
B. intermittent IV infusion is the best way to very rapidly achieve high drug concentrations
C. concentrations of drug in the body following multiple infusions are additive
D. all drugs are better tolerated as IV bolus doses compared to IV infusion

Question 8 A patient is to take a medication by mouth every 8 hours. The medication has an oral bioavailability of 83%, an absorption half-life of 90 minutes, total body clearance of 3.0 L/hr, and apparent volume of distribution of approximately 25 L. What is an appropriate dose to achieve an average plasma drug concentration of 25 mg/L in this patient? Give your answer in mg rounded to the nearest whole number with no units.

Question 9 What is the absolute bioavailability of a drug for which a 500 mg dose resulted in an AUC of 139 (mg/L)hr? The elimination half-life of the drug is 4.5 hr and the apparent volume of distribution is 18 L. Give your answer as a decimal rounded to the nearest hundredth with no units.

Question 10

Which term describes the comparison of the bioavailability of the active drug in the systemic circulation following extravascular administration with the bioavailability of the same drug following intravenous administration?

A. Drug Product Performance
B. Relative Bioavailability
C. Bioequivalence
D. Absolute Bioavailability

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working.

Question 1 Worked by the drill

Answer: A. to maintain plasma drug levels within the therapeutic window

One dose rises above the minimum effective concentration and then falls below it. Repeating the dose keeps the level between the minimum effective and the minimum toxic concentration, the therapeutic window.

Question 2 Worked by the drill

Answer: 151.1 mg/L

WHAT THE STEM GIVES

D_0	17 mg/kg	scaled to 75 kg: 1275 mg, the numerator of C_0
weight	75 kg	scales both the dose and the volume
τ	every 8 hours	the exponent $k\tau = 0.1733 \times 8 = 1.386$
t½	4 hours	gives $k = \frac{0.693}{4} = 0.1733\ \text{hr}^{-1}$
V_D	15% of body weight	scaled to 75 kg: 11.25 L, the denominator of C_0

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{max}^{\infty} = \frac{C_0}{1 - e^{-k\tau}}, \text{ with } C_0 = \frac{D_0}{V_D}$$

"IV bolus injections … every 8 hours" and "maximum steady-state concentration" name the repeated IV bolus block at steady state. The peak at the plateau is the single-dose peak divided by the fraction not yet eliminated at each dose.

UNIT

$$D_0 = 17\ \text{mg/kg} \times 75\ \text{kg} = 1275\ \text{mg};\ V_D = 0.15 \times 75\ \text{kg} = 11.25\ \text{L};\ C_0 = \frac{1275\ \text{mg}}{11.25\ \text{L}} = 113.3\ \text{mg/L}$$

Both the dose and the volume are given per kilogram of body weight, so both are scaled to the 75 kg patient before anything else. mg over L is mg/L.

ALGEBRA

$$k = \frac{0.693}{4\ \text{hr}} = 0.1733\ \text{hr}^{-1};\ e^{-k\tau} = e^{-0.1733 \times 8} = e^{-1.386} = 0.250$$

Eight hours is two half-lives, so one quarter of a dose remains when the next arrives. The exponent is dimensionless: reciprocal hours times hours.

ALGEBRA

$$C_{max}^{\infty} = \frac{113.3\ \text{mg/L}}{1 - 0.250} = \frac{113.3}{0.750} = 151.1\ \text{mg/L}$$

Dividing by 0.75 is the accumulation factor 1.33: the steady-state peak is one third higher than the first-dose peak.

ROUND

151.1

Her key is 151.1 with a margin of 5%, so 143.5 to 158.7 scores. One decimal place was asked for, and Canvas took the number with no unit.

> **DOES THE NUMBER MAKE SENSE?**
>
> C_0 is $\frac{17}{0.15}$ = 113.3 mg/L per dose and 8 hours is two half-lives, so a quarter is left at each new dose; the peak is C_0 over 0.750, a third more than 113.3: 151.1, between 113.3 and 2 × 113.3.

Question 3

Worked by the drill

Answer: A. decrease the dosing interval

Dosing more often puts the next dose in before as much of the last has left, so the plateau rises. Of the two things a regimen can change, the dose and the interval, this is the interval change that raises the level.

Question 4

Worked by the drill

Answer: 10.6 mg/L

WHAT THE STEM GIVES

D_0	Four hundred milligrams	with the infusion time gives R = $\frac{400}{2}$ = 200 mg/hr
infusion time	over a period of 2 hours	R = 200 mg/hr; t = 2 hr in $1 - e^{-kt}$
second start	Eight hours after the start of the first infusion	the second infusion runs 8 to 10 hr
t½	approximately 4 hours	gives k = $\frac{0.693}{4}$ = 0.1733 hr^{-1}
V_D	approximately 20 L	in Cl = kV_D = 3.465 L/hr
t	4 hours after the cessation of the second infusion	the second term decays 4 hr, the first 12 hr

THE WORKING, ONE LINE AT A TIME

SETUP

Each infusion ends at Cend = $\frac{R}{k \times V_D}(1 - e^{-k \times 2\ hr})$; then C_p = Cend $e^{-k\ t_1}$ + Cend $e^{-k\ t_2}$, one term per infusion

The two infusions are added (superposition). Each is a 2-hour infusion that ends at the same concentration because the dose, the time and the patient are the same; after it stops, each decays by first-order elimination from its own end time.

UNIT

$R = \frac{400\ mg}{2\ hr} = 200$ mg/hr; $k = \frac{0.693}{4\ hr} = 0.1733\ hr^{-1}$; $Cl = k \times V_D = 0.1733 \times 20$ L = 3.465 L/hr

The rate is the dose over the infusion time, in mg/hr. Clearance comes from k and V_D because the infusion line wants Cl.

ALGEBRA

Cend = $\frac{200\ mg/hr}{3.465\ L/hr}(1 - e^{-0.1733 \times 2}) = 57.72 \times (1 - 0.7071) = 16.90$ mg/L

57.72 mg/L would be the plateau if the drip ran on; after two hours, half a half-life, it has reached 29.3% of that.

ALGEBRA

Times since each infusion ended: first ended at 2 hr, so 12 hr have passed at t = 14 hr; second ended at 10 hr, so 4 hr. $C_p = 16.90\ e^{-0.1733 \times 12} + 16.90\ e^{-0.1733 \times 4} = 16.90 \times 0.125 + 16.90 \times 0.500 = 2.11 + 8.45 = 10.56$ mg/L

The second infusion started 8 hours after the first began, so it ran from 8 to 10 hours; 4 hours after its end is 14 hours. Twelve hours is three half-lives (one eighth left), four hours is one half-life (one half left).

ROUND

10.6

Her key is 10.6 with a margin of 5%, so 10.1 to 11.1 scores. Leaving out the first infusion gives 8.45, too low; treating each infusion as a 400 mg bolus into 20 L gives 20 × 0.125 + 20 × 0.500 = 12.5, too high, because each 2-hour infusion ends at 16.90 mg/L.

DOES THE NUMBER MAKE SENSE?

Each infusion ends at 16.90 mg/L. Four hours is one half-life, so the second leaves 8.45; the first, 12 hours or three half-lives on, leaves an eighth, 2.11. The sum is between 8.45 and 2 × 16.90.

Question 5

Worked by the drill

Answer: D. Increased steady-state C_{max}

Every concentration in the regimen scales with the dose, so the steady-state peak rises in proportion. Slide "Altering Dose": increasing the dose increases steady-state concentrations and the peak-to-trough fluctuation.

Question 6

Worked by the drill

Answer: 2.6 hr

WHAT THE STEM GIVES

weight	74-kg	*not needed: t_{max}^{∞} has no volume in it*
D_0	750 mg	*not needed: t_{max}^{∞} does not depend on the dose*
τ	every 8 hours	in $1 - e^{-k\tau}$ and $1 - e^{-k_a\tau}$
F	92%	*not needed: F scales the height of the curve, not its timing*
V_D	0.29 L/kg	*not needed*
t½	5 hours	gives $k = \frac{0.693}{5} = 0.1386\ hr^{-1}$
absorption t½	1.5 hour	gives $k_a = \frac{0.693}{1.5} = 0.462\ hr^{-1}$

THE WORKING, ONE LINE AT A TIME

SETUP

$$t_{max}^{\infty} = \frac{1}{k_a - k}\ \ln\left[\frac{k_a(1 - e^{-k\tau})}{k(1 - e^{-k_a\tau})}\right]$$

"Oral dose ... every 8 hours" and "when does the ... maximum concentration ... at steady-state occur" ask for the time of the peak within an interval at the plateau. Only the two rate constants and τ enter; the dose, F, V_D and the weight are not needed.

ALGEBRA

$$k = \frac{0.693}{5\ hr} = 0.1386\ hr^{-1};\ k_a = \frac{0.693}{1.5\ hr} = 0.462\ hr^{-1}$$

Each half-life gives its own rate constant. The absorption half-life is the shorter one, so k_a is the larger constant, as it must be for an immediate-release product.

ALGEBRA

$e^{-k\tau} = e^{-0.1386 \times 8} = 0.330$, so $1 - 0.330 = 0.670$; $e^{-k_a\tau} = e^{-0.462 \times 8} = 0.0248$, so $1 - 0.0248 = 0.975$

Both accumulation terms are evaluated at τ = 8 hours. Absorption is nearly complete within an interval (0.975), elimination is not (0.670).

ALGEBRA

$$t_{max}^{\infty} = \frac{1}{0.462 - 0.1386}\ \ln\left[\frac{0.462 \times 0.670}{0.1386 \times 0.975}\right] = \frac{1}{0.3234}\ \ln(2.291) = 3.092 \times 0.829 = 2.56\ hr$$

The ratio inside the log is 2.291 and its natural log is 0.829; dividing by the difference of the constants gives hours. The single-dose t_{max} for the same constants would be 3.72 hr, so the steady-state peak comes earlier.

ROUND

2.6

Her key is 2.6 with a margin of 5%, so 2.5 to 2.7 scores. The single-dose line, $\frac{\ln(k_a \div k)}{k_a - k}$, gives 3.72 hr with the same constants; at steady state the peak comes earlier, so the τ terms are needed.

DOES THE NUMBER MAKE SENSE?

The steady-state peak must fall inside the 8-hour interval and comes earlier than a first-dose peak, since drug from earlier doses is being eliminated while the new dose absorbs: 2.6 hours, between 0 and 8.

Question 7 Worked by the drill

Answer: C. concentrations of drug in the body following multiple infusions are additive

This is superposition applied to infusions: each infusion is worked as a single infusion, decays from its own end, and the concentrations are added. It is the assumption behind every two-infusion calculation.

Question 8 Worked by the drill

Answer: 723 mg

WHAT THE STEM GIVES

τ	every 8 hours	multiplies clearance: Cl × τ = 3.0 × 8 = 24 L
F	83%	written 0.83, the denominator
absorption t½	90 minutes	*not needed: the average concentration does not depend on k_a*
Cl	3.0 L/hr	in the numerator with τ
V_D	approximately 25 L	*not needed: clearance is given*
C_{avg}^{∞}	25 mg/L	the target, in the numerator

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{avg}^{\infty} = \frac{F \times D_0}{Cl \times \tau}, \text{ so } D_0 = \frac{C_{avg}^{\infty} \times Cl \times \tau}{F}$$

"By mouth every 8 hours" and "average plasma drug concentration" name the average steady-state line of the multiple oral block, rearranged for the dose. Clearance is given directly, so k and V_D are not needed; the absorption half-life and V_D are extra data for this part.

ALGEBRA

$$D_0 = \frac{25 \text{ mg/L} \times 3.0 \text{ L/hr} \times 8 \text{ hr}}{0.83} = \frac{600 \text{ mg}}{0.83} = 722.9 \text{ mg}$$

mg/L × L/hr × hr leaves mg: the amount cleared in one interval at the target average. Dividing by F scales it up to the dose that has to be swallowed for 83% of it to reach the blood.

ROUND

723

Her key is 723 with a margin of 5%, so 687 to 759 scores. Multiplying by F instead of dividing gives 600 × 0.83 = 498 mg, less than the 600 mg that has to be absorbed, which cannot be right. The whole number was asked for.

DOES THE NUMBER MAKE SENSE?

Holding 25 mg/L while 24 L is cleared per interval needs 600 mg absorbed; with only 0.83 of the dose absorbed, the dose must be larger than 600 mg: $\frac{600}{0.83}$ = 723 mg.

Question 9

Worked by the drill

Answer: 0.77 (as a decimal fraction)

WHAT THE STEM GIVES

D_0	500 mg	the numerator of $AUC_{IV} = \frac{D_{IV}}{Cl}$; the dose ratio is 1
AUC_{po}	139 (mg/L)hr	the numerator of the area ratio
t½	4.5 hr	gives $k = \frac{0.693}{4.5} = 0.154\ hr^{-1}$
V_D	18 L	in $Cl = kV_D = 0.154 \times 18 = 2.772$ L/hr

THE WORKING, ONE LINE AT A TIME

SETUP

$$F_{abs} = \frac{AUC_{po}}{AUC_{IV}} \text{ with the same dose, and } AUC_{IV} = \frac{D_{IV}}{Cl},\ Cl = k \times V_D$$

"Absolute bioavailability" with one AUC given and no IV AUC: the IV area has to come from the dose and the clearance, which the half-life and the volume supply. With equal doses the dose ratio is 1.

ALGEBRA

$$k = \frac{0.693}{4.5\ hr} = 0.154\ hr^{-1};\ Cl = 0.154\ hr^{-1} \times 18\ L = 2.772\ L/hr;\ AUC_{IV} = \frac{500\ mg}{2.772\ L/hr} = 180.4\ (mg/L)hr$$

$hr^{-1} \times L$ is L/hr, and mg over L/hr is mg·hr/L, the unit of an AUC. This is the area the same 500 mg would give intravenously.

ALGEBRA

$$F_{abs} = \frac{139}{180.4} = 0.771$$

With equal doses the dose ratio is $\frac{500}{500} = 1$ and drops out, leaving $F_{abs} = \frac{AUC_{po}}{AUC_{IV}} = \frac{139}{180.4} =$ 0.771, a dimensionless fraction. The oral area is 77 per cent of the IV area, so 77 per cent of the oral dose reached the circulation.

ROUND

0.77

Her key is 0.77 with a margin of 3%, so 0.75 to 0.79 scores. Two decimal places were asked for, as a decimal, not a percentage.

DOES THE NUMBER MAKE SENSE?

Clearance 2.772 L/hr means the 500 mg would give 180.4 (mg/L)hr if all of it reached the blood; the oral 139 is about three-quarters of that, so F is about 0.77, between 0 and 1.

Question 10 Worked by the drill

Answer: D. Absolute Bioavailability

The stem is the slide definition word for word. Absolute means measured against the intravenous dose, where all of the drug is in the circulation.
